

ISEP INSTITUTO SUPERIOR DE ENGENHARIA DO PORTO

HTML



Sistemas de Informação 1

Content

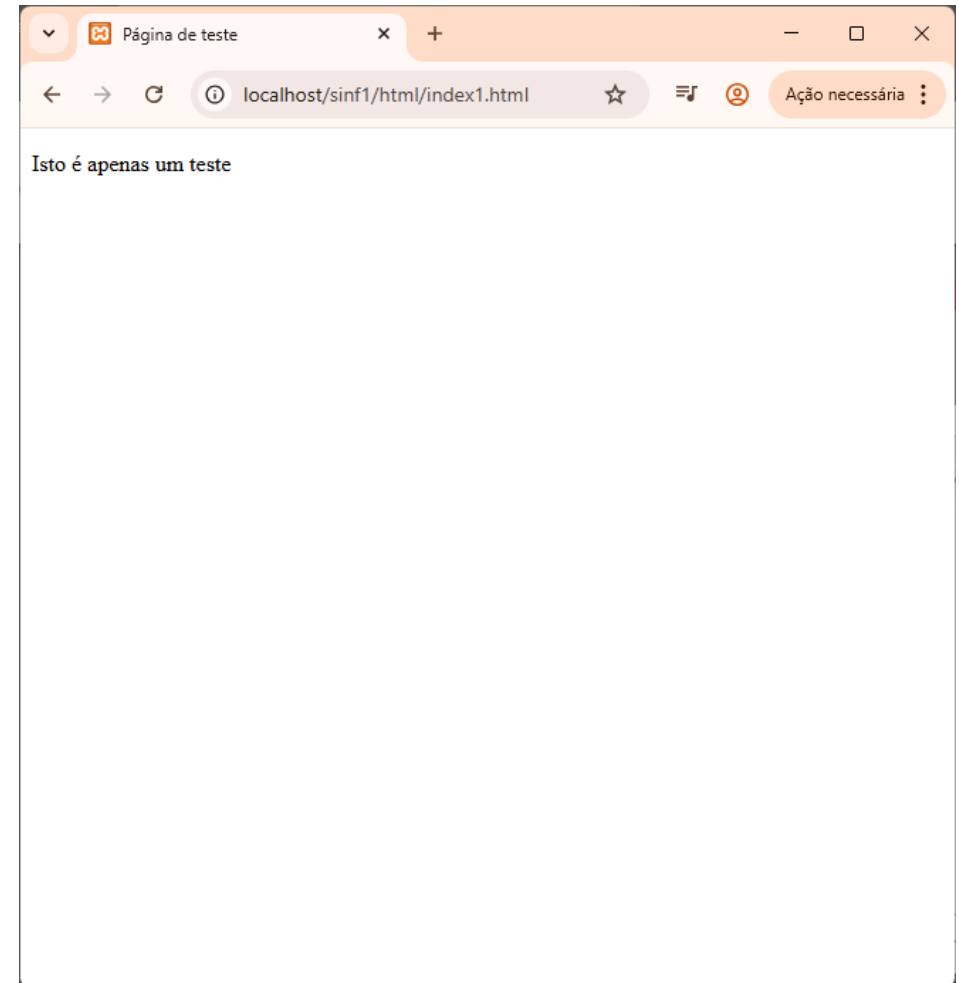
1. Context
2. Simple example
3. Headers and paragraphs
4. Formatting
5. Images and links
6. Lists
7. Tables
8. HTML 5: Semantic elements
9. Forms
10. Forms – elements
11. HTML 5 Forms
12. Media
13. Best Practices

Context

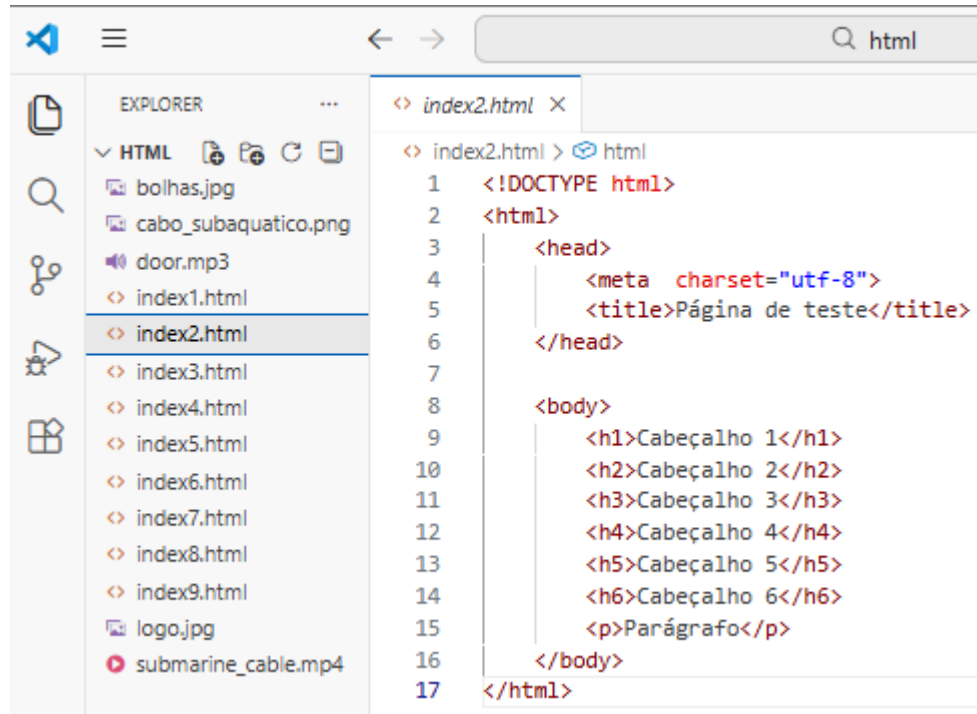
HTML: HyperText Markup Language

- Created by Tim Berners-Lee
- Used to describe and define the content in a web page
- Maintained by the W3C consortium
- Text **markup** language (**not a programming language**)
 - Markup used to mark text, images and other content to be shown in a browser
- **Hipertext** or **hiperlinks** (links): allow the connection between several pages
- Tutorial, element list and examples at: <https://www.w3schools.com/html/>

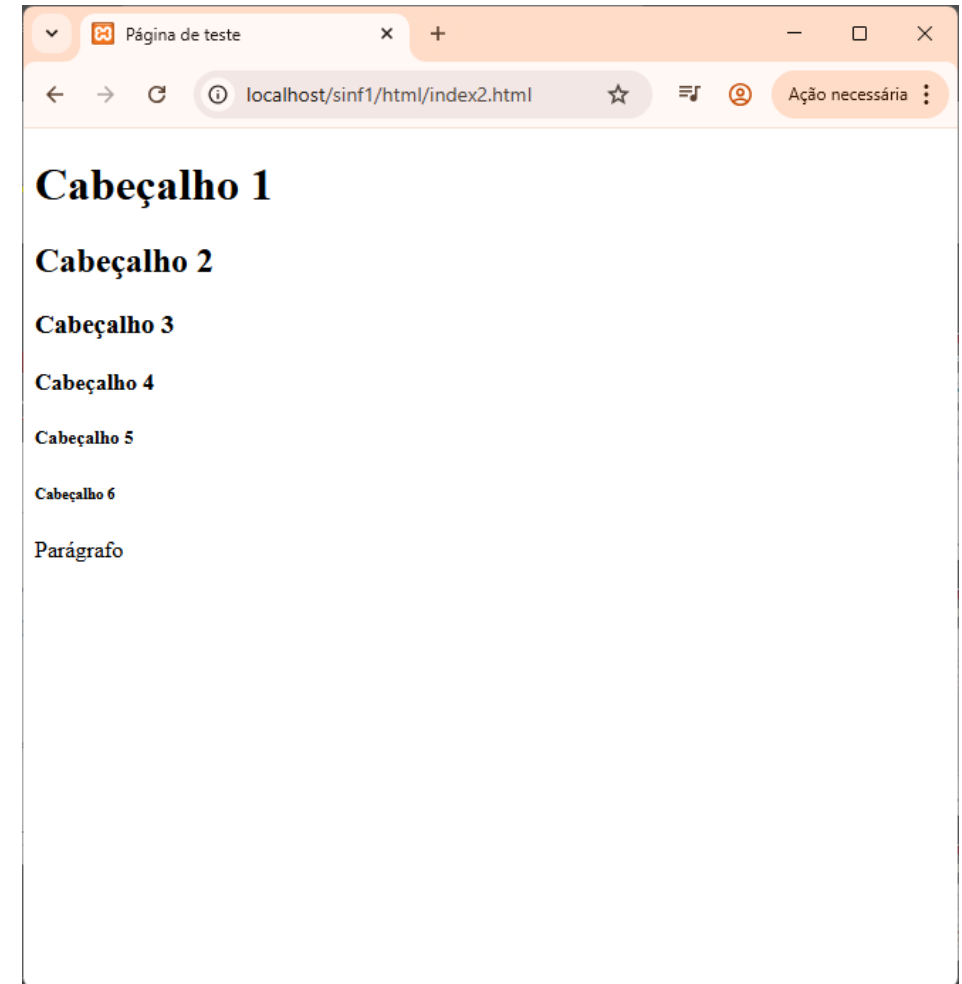
Simple example



Headers and Paragraphs



```
<!-- index2.html -->
<!DOCTYPE html>
<html>
  <head>
    <meta charset="utf-8">
    <title>Página de teste</title>
  </head>
  <body>
    <h1>Cabeçalho 1</h1>
    <h2>Cabeçalho 2</h2>
    <h3>Cabeçalho 3</h3>
    <h4>Cabeçalho 4</h4>
    <h5>Cabeçalho 5</h5>
    <h6>Cabeçalho 6</h6>
    <p>Parágrafo</p>
  </body>
</html>
```

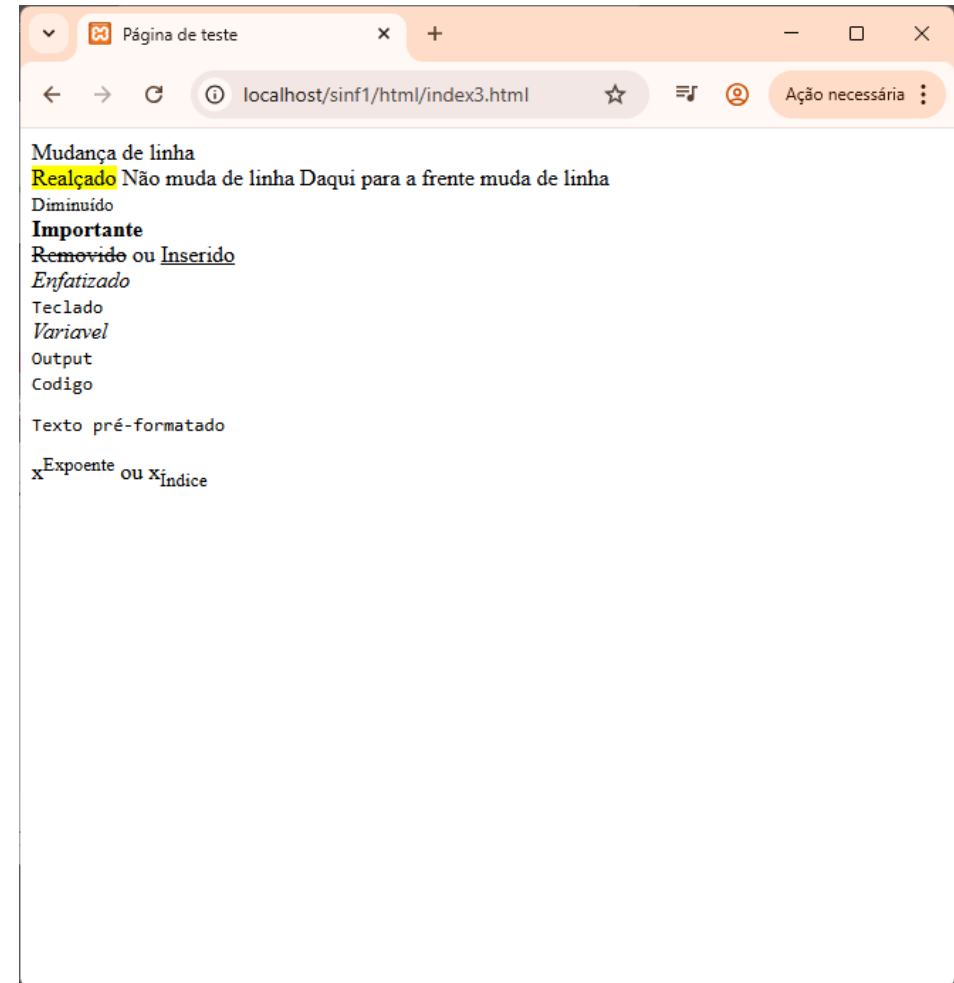


Formatting

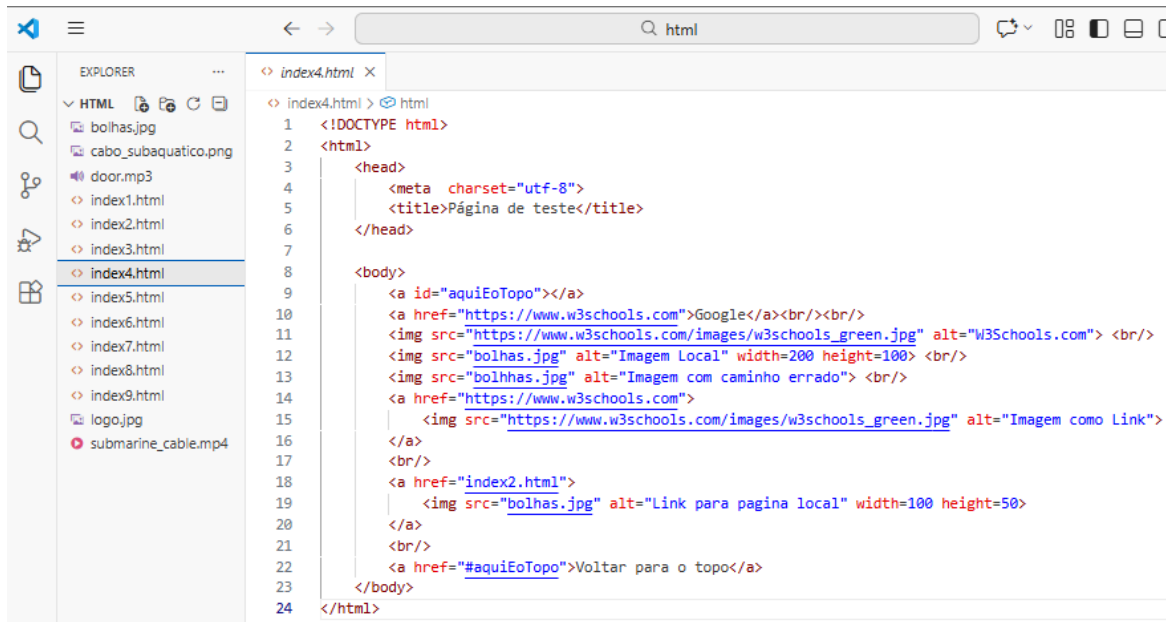
```

1  <!DOCTYPE html>
2  <html>
3      <head>
4          <meta charset="utf-8">
5          <title>Página de teste</title>
6      </head>
7
8      <body>
9          Mudança de linha <br/>
10         <mark>Realçado</mark> Não muda de linha
11         Daqui para a frente muda de linha<br/>
12         <small>Diminuído</small><br/>
13         <strong>Importante</strong><br/>
14         <del>Removido</del> ou <ins>Inserido</ins><br/>
15         <em>Enfatizado</em><br/>
16         <kbd>Teclado</kbd><br/>
17         <var>Variavel</var><br/>
18         <samp>Output</samp><br/>
19         <code>Codigo</code><br/>
20         <pre>Texto pré-formatado </pre>
21         x<sup>Expoente</sup> ou x<sub>Índice</sub>
22     </body>
23 </html>

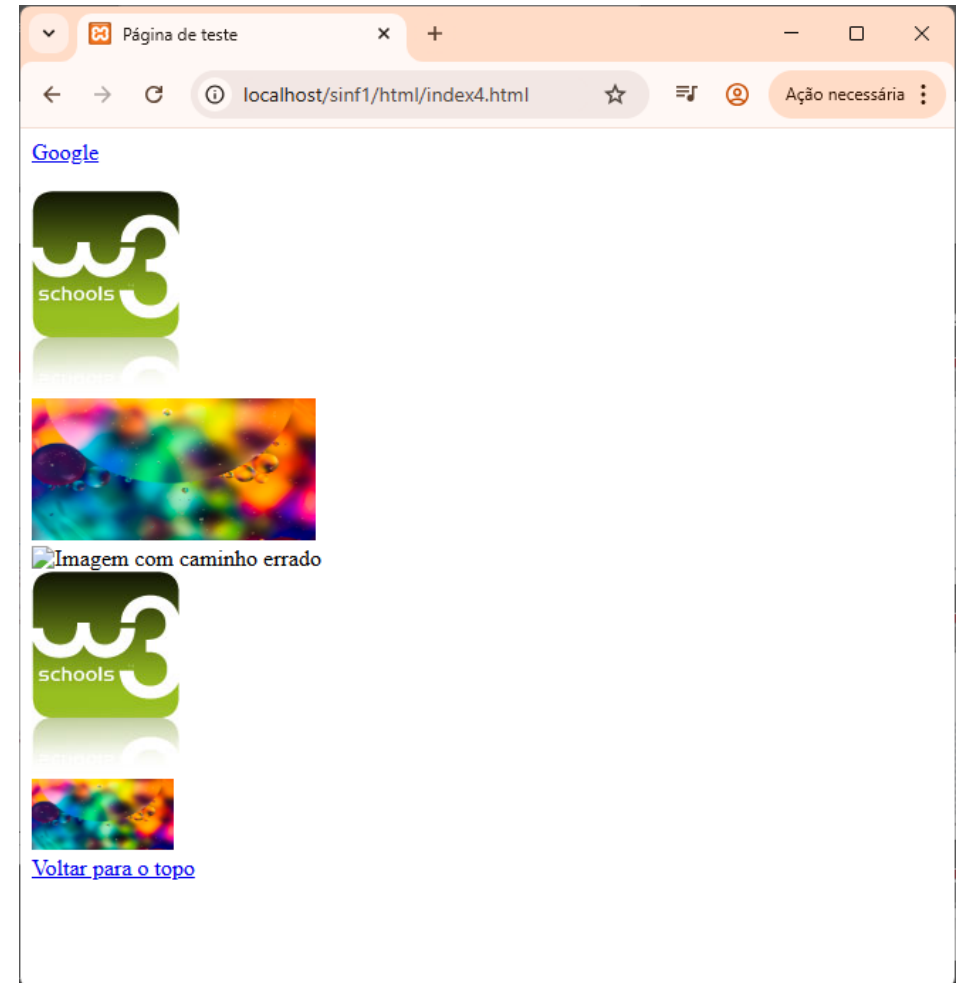
```



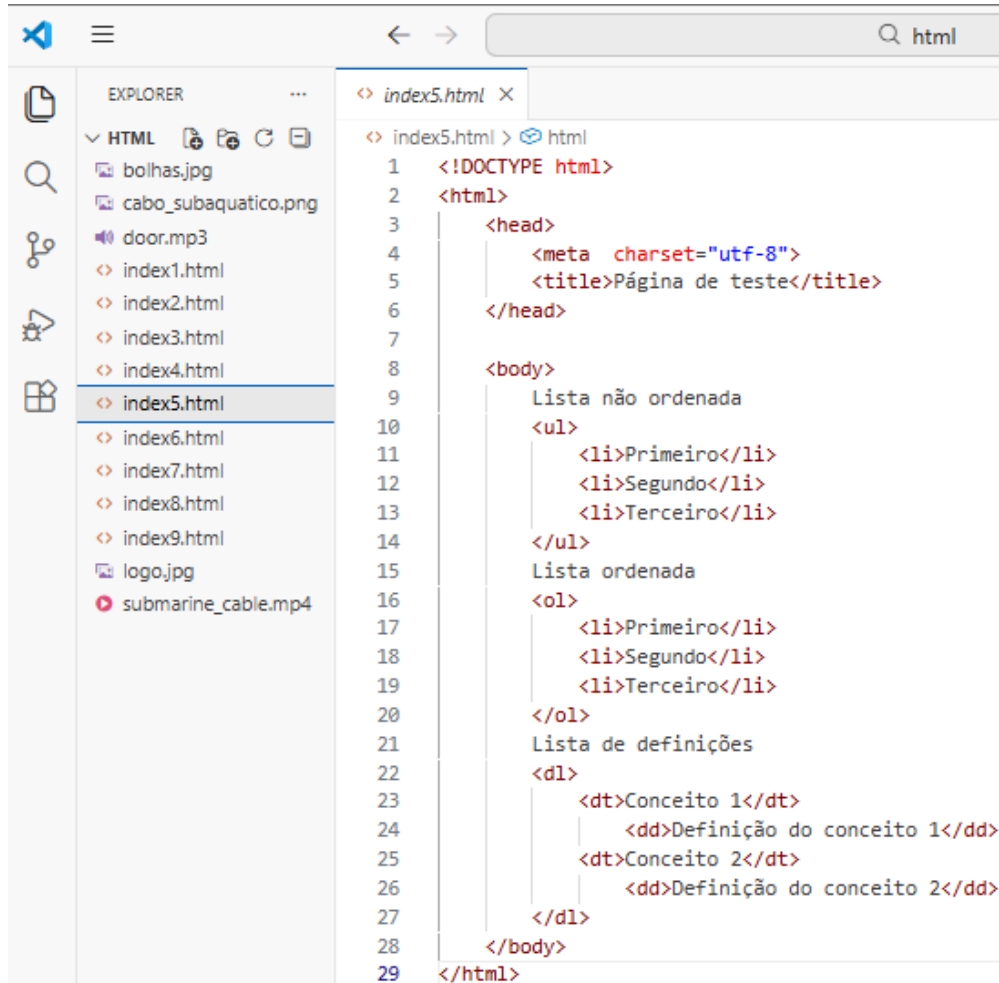
Images and links



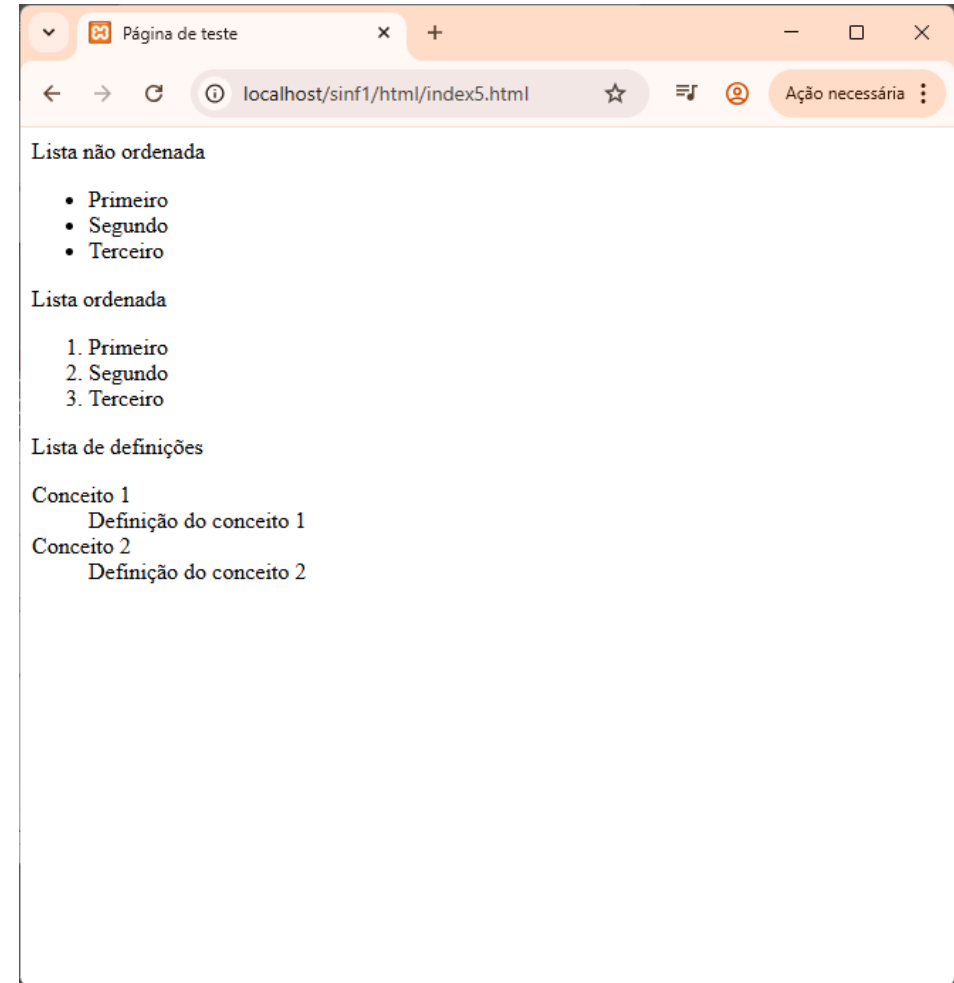
```
1 <!DOCTYPE html>
2 <html>
3   <head>
4     <meta charset="utf-8">
5     <title>Página de teste</title>
6   </head>
7   <body>
8     <a id="aquiEoTopo"></a>
9     <a href="https://www.w3schools.com">Google</a><br/><br/>
10     <br/>
11     <br/>
12     <br/>
13    <a href="https://www.w3schools.com">
14      
15    </a>
16    <br/>
17    <a href="index2.html">
18      
19    </a>
20    <br/>
21    <a href="#aquiEoTopo">Voltar para o topo</a>
22  </body>
23 </html>
```



Lists



```
1 <!DOCTYPE html>
2 <html>
3   <head>
4     <meta charset="utf-8">
5     <title>Página de teste</title>
6   </head>
7
8   <body>
9     Lista não ordenada
10    <ul>
11      <li>Primeiro</li>
12      <li>Segundo</li>
13      <li>Terceiro</li>
14    </ul>
15    Lista ordenada
16    <ol>
17      <li>Primeiro</li>
18      <li>Segundo</li>
19      <li>Terceiro</li>
20    </ol>
21    Lista de definições
22    <dl>
23      <dt>Conceito 1</dt>
24      <dd>Definição do conceito 1</dd>
25      <dt>Conceito 2</dt>
26      <dd>Definição do conceito 2</dd>
27    </dl>
28  </body>
29 </html>
```



Tables

```

1 <!DOCTYPE html>
2 <html>
3   <head>
4     <meta charset="utf-8">
5     <title>Página de teste</title>
6   </head>
7   <body>
8     <table border="1">
9       <tr><th>Cabeçalho Coluna 1</th><th>Cabeçalho Coluna 2</th><th>Cabeçalho Coluna 3</th></tr>
10      <tr><td>Conteúdo 1-1</td><td>Conteúdo 1-2</td><td>Conteúdo 1-3</td></tr>
11      <tr><td>Conteúdo 2-1</td><td>Conteúdo 2-2</td><td>Conteúdo 2-3</td></tr>
12      <tr><td>Conteúdo 3-1</td><td>Conteúdo 3-2</td><td>Conteúdo 3-3</td></tr>
13      <tr><td>Conteúdo 4-1</td><td>Conteúdo 4-2</td><td>Conteúdo 4-3</td></tr>
14    </table>
15
16    <br>
17
18    <table border="1">
19      <tr><th>Cabeçalho Coluna 1</th><th>Cabeçalho Coluna 2</th><th>Cabeçalho Coluna 3</th></tr>
20      <tr><td>Conteúdo 1-1</td><td>Conteúdo 1-2</td><td>Conteúdo 1-3</td></tr>
21      <tr><td colspan="2" align="center">Conteúdo 2-1 e 2-2</td><td>Conteúdo 2-3</td></tr>
22      <tr><td>Conteúdo 3-1</td><td>Conteúdo 3-2</td><td rowspan="2">Conteúdo 3-3 e 4-3</td></tr>
23      <tr><td>Conteúdo 4-1</td><td>Conteúdo 4-2</td></tr>
24    </table>
25  </body>
26 </html>

```

Página de teste

localhost/sinf1/html/index6.html

Cabeçalho Coluna 1	Cabeçalho Coluna 2	Cabeçalho Coluna 3
Conteúdo 1-1	Conteúdo 1-2	Conteúdo 1-3
Conteúdo 2-1	Conteúdo 2-2	Conteúdo 2-3
Conteúdo 3-1	Conteúdo 3-2	Conteúdo 3-3
Conteúdo 4-1	Conteúdo 4-2	Conteúdo 4-3

Cabeçalho Coluna 1	Cabeçalho Coluna 2	Cabeçalho Coluna 3
Conteúdo 1-1	Conteúdo 1-2	Conteúdo 1-3
Conteúdo 2-1 e 2-2		Conteúdo 2-3
Conteúdo 3-1	Conteúdo 3-2	Conteúdo 3-3 e 4-3
Conteúdo 4-1	Conteúdo 4-2	

HTML5 Semantic elements

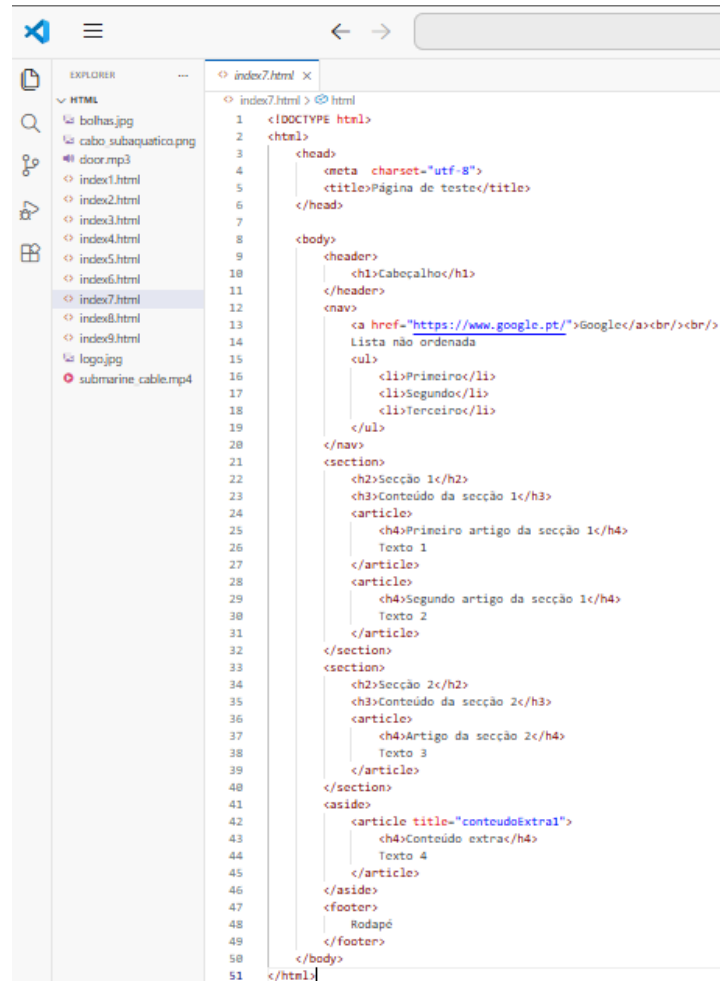


Objective: besides structuring documents, also describe the documents' meanings

Elements:

- **<header>** document or section header. May contain the elements <h1>...<h6>
- **<nav>** groups of links. May contain <a>, e
- **<section>** document section. Contains a title (<h1>...<h6>).
- **<article>** title identification recommended
- **<aside>** additional support content
- **<footer>** (author, contacts, date, ...)

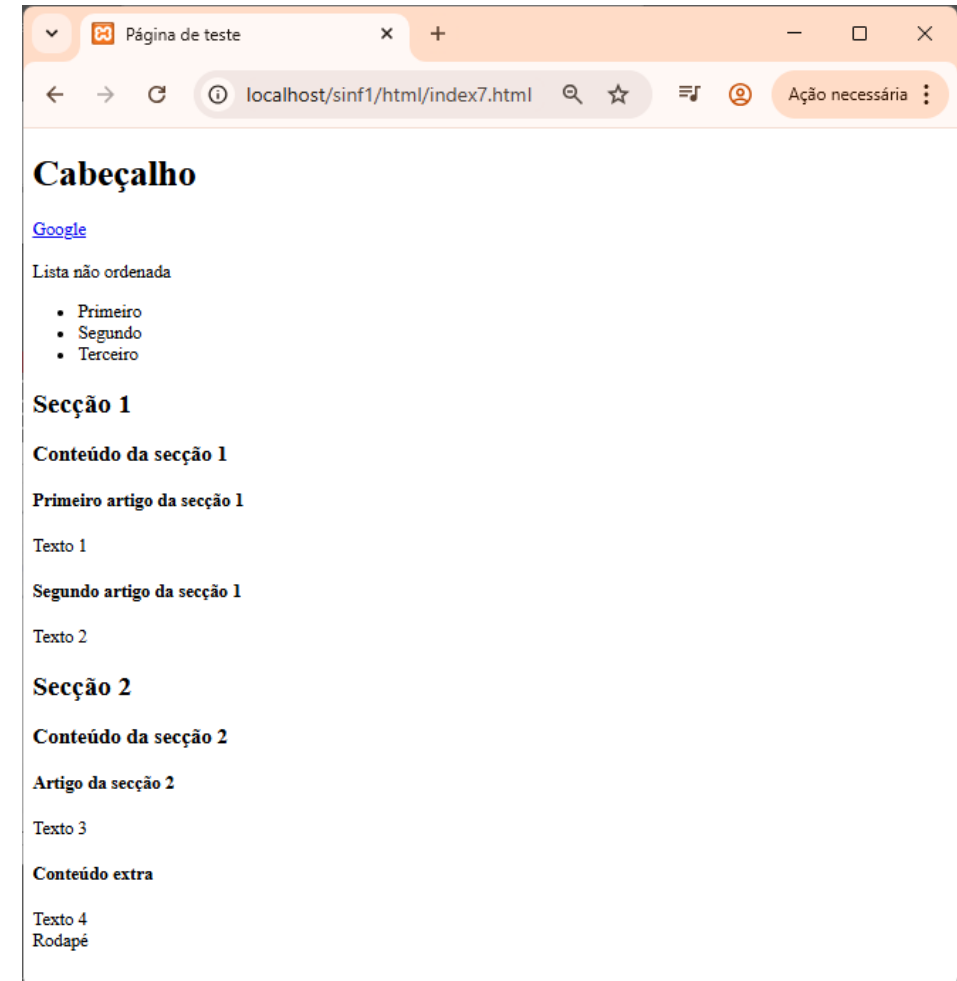
HTML5 Semantic elements



```

1 <!DOCTYPE html>
2 <html>
3   <head>
4     <meta charset="utf-8">
5     <title>Página de teste</title>
6   </head>
7   <body>
8     <header>
9       <h1>Cabeçalho</h1>
10    </header>
11    <nav>
12      <a href="https://www.google.pt/">Google</a><br/><br/>
13      Lista não ordenada
14      <ul>
15        <li>Primeiro</li>
16        <li>Segundo</li>
17        <li>Terceiro</li>
18      </ul>
19    </nav>
20    <section>
21      <h2>Secção 1</h2>
22      <h3>Conteúdo da secção 1</h3>
23      <article>
24        <h4>Primeiro artigo da secção 1</h4>
25        Texto 1
26      </article>
27      <article>
28        <h4>Segundo artigo da secção 1</h4>
29        Texto 2
30      </article>
31    </section>
32    <section>
33      <h2>Secção 2</h2>
34      <h3>Conteúdo da secção 2</h3>
35      <article>
36        <h4>Artigo da secção 2</h4>
37        Texto 3
38      </article>
39    </section>
40    <aside>
41      <article title="conteúdoExtra1">
42        <h4>Conteúdo extra</h4>
43        Texto 4
44      </article>
45    </aside>
46    <footer>
47      Rodapé
48    </footer>
49  </body>
50 </html>

```



Forms

```

1 <!DOCTYPE html>
2 <html>
3   <head>
4     <meta charset="utf-8">
5     <title>Página de teste</title>
6   </head>
7
8   <body>
9     <form action="index7.html" method="post">
10      <fieldset>
11        <legend>Dados Pessoais:</legend>
12        <label>Email:</label><br/>
13        <input name="email" type="email"><br/>
14        <label>Password:</label><br/>
15        <input name="password" type="password"><br/>
16      </fieldset>
17      <fieldset>
18        <legend>Questionário</legend>
19        <label>Comentário:</label><br/>
20        <textarea name="comentario"></textarea><br/>
21        <label>Quando deseja:</label><br/>
22        <select>
23          <option value="ontem">Ontem</option>
24          <option value="hoje">Hoje</option>
25          <option value="amanha">Amanhã</option>
26        </select>
27        <br/>
28        <label>Que meio(s) de transporte costuma usar:</label><br/>
29
30        <input type="checkbox" name="vehicle1" value="Bike">Bicicleta<br/>
31        <input type="checkbox" name="vehicle2" value="Car">Automóvel<br/>
32        <input type="checkbox" name="vehicle3" value="Public" checked="">Transportes Públicos<br/>
33
34        <label>Quanto tempo demora:</label><br/>
35        <input type="radio" name="tempo" value="pouco"> Menos de 30 min<br/>
36        <input type="radio" name="tempo" value="muito"> Mais de 30 min<br/>
37
38        <button type="button">Info</button>
39        <button type="reset">Limpar</button>
40        <button type="submit">Enviar</button>
41      </fieldset>
42    </form>
43  </body>
44 </html>

```

Página de teste

localhost/sinf1/html/index8.html

Ação necessária

Dados Pessoais:

Email:

Password:

Questionário

Comentário:

Quando deseja:

Ontem

Que meio(s) de transporte costuma usar?

☐ Bicicleta

☐ Automóvel

☒ Transportes Públicos

Quanto tempo demora?

☐ Menos de 30 min

☐ Mais de 30 min

Info Limpar Enviar

Forms - elements

■ **form:**

- **action:** where the info goes after submission
- **method:** submission method. Possible values: POST, GET

GET	POST
Appends the parameters and values to the URL	Appends the parameters and values to the HTTP request body
URL size is limited (3000)	No URL limit size
Allows saving an URL with parameters	Allows saving the URL without parameters

■ **fieldset:** container to group elements

■ **legend:** legend for the **fieldset** element

■ **label**

■ **input:** data input field

- type: text | email | password | checkbox | radio

■ **textarea:** large text input field

■ **select:** drop-down list

- option: drop-down options

■ **button**

- type: button | reset | submit

HTML5 Forms

Values for the **type** attribute:

- color
- date
- datetime-local
- email
- month
- number
- range
- search
- tel
- time
- url
- week

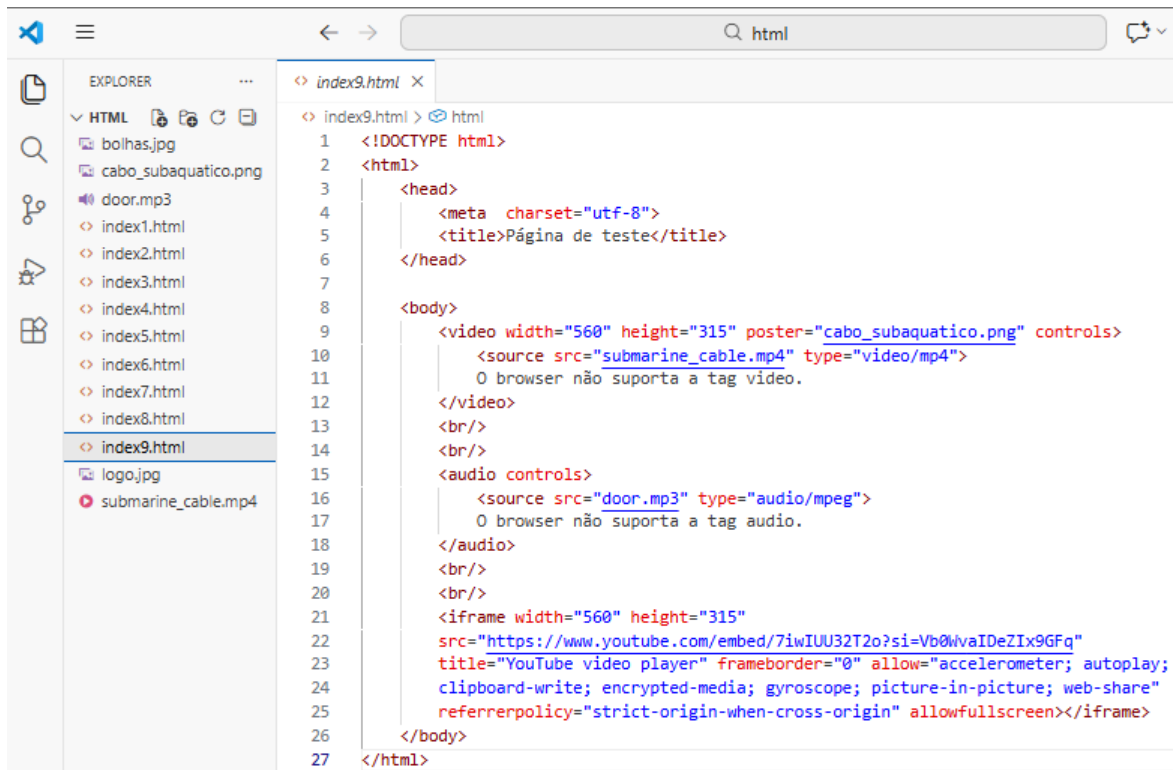
https://www.w3schools.com/html/html_form_input_types.asp

Attributes for the **input** tag:

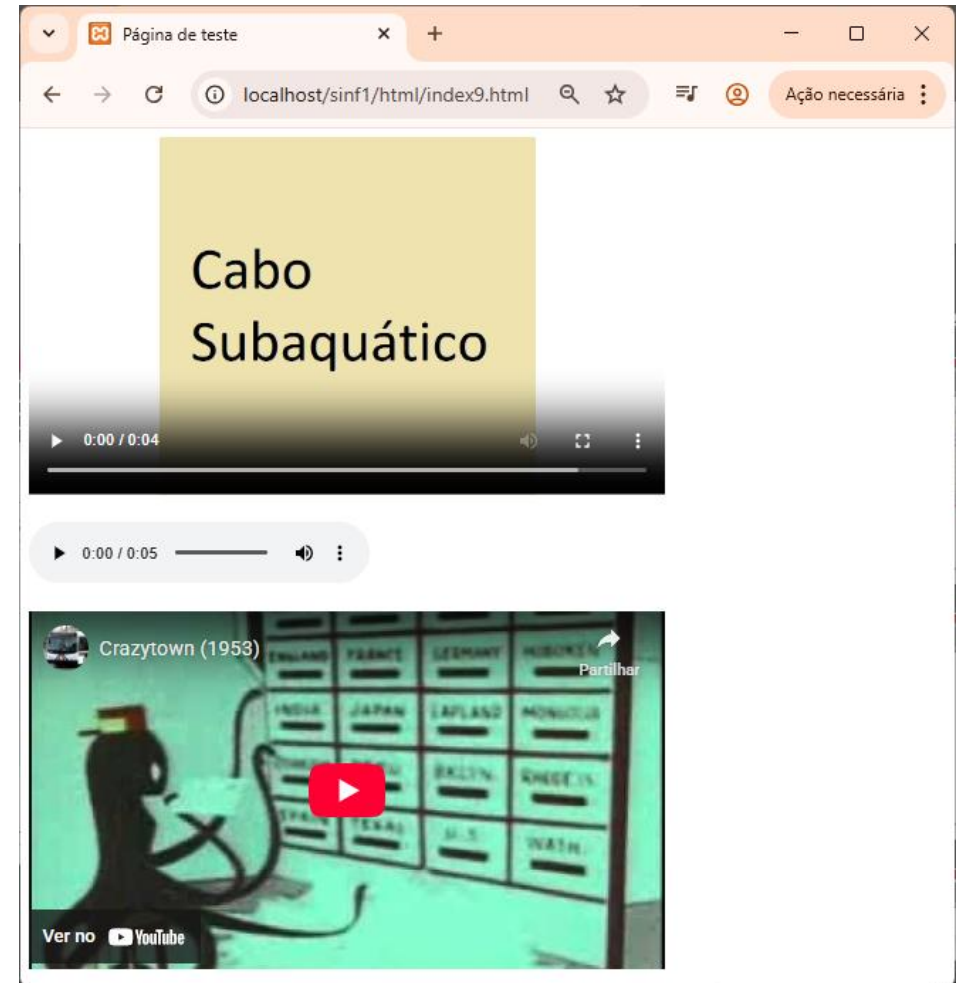
- autocomplete
- autofocus
- form
- formaction
- formenctype
- formmethod
- formnovalidate
- formtarget
- height and width
- list
- min and max
- multiple
- pattern (regexp)
- placeholder
- required
- step

https://www.w3schools.com/html/html_form_attributes.asp

Media



```
1 <!DOCTYPE html>
2 <html>
3   <head>
4     <meta charset="utf-8">
5     <title>Página de teste</title>
6   </head>
7
8   <body>
9     <video width="560" height="315" poster="cabo_subaquatico.png" controls>
10       <source src="submarine_cable.mp4" type="video/mp4">
11       0 browser não suporta a tag video.
12     </video>
13     <br/>
14     <br/>
15     <audio controls>
16       <source src="door.mp3" type="audio/mpeg">
17       0 browser não suporta a tag audio.
18     </audio>
19     <br/>
20     <br/>
21     <iframe width="560" height="315"
22       src="https://www.youtube.com/embed/7iwIUU32T2o?si=Vb0WvaIDeZIx9GFq"
23       title="YouTube video player" frameborder="0" allow="accelerometer; autoplay;
24       clipboard-write; encrypted-media; gyroscope; picture-in-picture; web-share"
25       referrerpolicy="strict-origin-when-cross-origin" allowfullscreen></iframe>
26   </body>
27 </html>
```



Best practices

General

- Clearly document all code using `<!--comment-->`. The code is more readable.
- Use the global attribute `title`, containing text that represents information related to the element it belongs to

Semantic elements

- Use HTML5 semantic elements so that the content is clearer to programmers, browsers and software that automatically process the information
- Group and organise the content through the semantic elements

Media

- Use the elements `video` and `audio`

Best practices (cont.)

Forms

- Use client-side validation as much as possible, in order to not overload the server
- When validating input, use the values of the attribute type
- Use the attributes of element input to simplify data validation
- Use the element label to define input labels
- Simplify the data insertion by using the attribute placeholder, that specifies a “tip”
- Make sure the data are being sent to the server the right way by defining the attribute method